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Haofei Yu

Research focus: user simulation and interaction; auto-research agents; multi-agent and agentic RL.

Education

2024 – present, Urbana, IL — **Ph.D.** in Computer Science, University of Illinois Urbana-Champaign
GPA: 3.96/4.00. Advisor: Prof. Jiaxuan You. Expected 2027.

2022 – 2024, Pittsburgh, PA — **M.S.** in Intelligent Information Systems, Carnegie Mellon University
GPA: 4.12/4.33. Advisor: Prof. Louis-Philippe Morency.

2018 – 2022, Hangzhou, China — **B.Eng.** in Computer Science, Zhejiang University
GPA: 3.96/4.00. Rank 7/134. Outstanding Graduate with Honors.

Experience

May – Aug 2026 — **Meta AI** — *Research Intern*
- Recast agentic continual learning as Bayesian discovery [12]; built DiscoverCL, a training-free meta-harness that keeps falsifiable hypotheses with evidence, and DiscoverScore, a meta-evaluation of how many bits each task teaches.
- On six CL-Bench streams: best bounded-memory method in 9/12 settings; recovers 50–97% of full-context ICL's gain with 2.6–9.6× less memory.

May – Aug 2025 — **Allen Institute for AI** — *Research Intern*
- Built ArtifactLinker [5] (COLM 2026) for rank-then-verify SOTA discovery.
- GNN ranking recovers 50% of known SOTA results in ∼10 verifications (vs. 60+); agentic verification reaches 72.6% success on 263 curated model–dataset pairs.

May – Aug 2024 — **MIT-IBM Watson AI Lab** — *Research Intern*
- Built a lightweight SWE-bench agent harness with fine-grained diagnostics.

May – Aug 2023 — **Apple AIML** — *Machine Learning Intern*
- Built an LLM-based hierarchical prompting system for Siri query disambiguation, improving internal user satisfaction; presented to Siri leadership.

Feb – Aug 2022 — **Tencent AI Lab** — *Research Intern*
- Training-free memory selection for long-context LMs (Findings of EMNLP 2023).

Research Highlights

User Simulation and Interaction Agents

- **Social World Models** [1] learn belief transitions by optimizing an ELBO on market time series. They achieve state-of-the-art Kalshi forecasts on SWM-bench, which contains 12K+ observations.
- **SOTOPIA-RL** [2] trains social agents with utterance-level credit assignment and multidimensional rewards. This yields goal-completion scores of 7.17 on SOTOPIA-hard and 8.31 on SOTOPIA-full.
- **SOTOPIA-π** [3] trains social agents through behavior cloning and self-reinforcement on LLM-filtered dialogues. This enables a 7B model to match GPT-4 on social goal completion.
- **macro2mind** [4] learns behavioral reasoning from markets using GRPO and a hindsight-regret curriculum. It transfers zero-shot to four user-simulation benchmarks and generates data that improves a downstream simulator's unseen-user accuracy by 15.5 percentage points.

Auto-Research Agents and Benchmarks

- **TinyScientist** [6] lets users guide research agents by editing intermediate outputs. On 20 ML tasks, its papers score ~0.5 points higher than Agent Laboratory's in human ratings on a 1–5 scale.
- **ResearchTown** [7] models research communities as researcher–paper graphs using TextGNN. Its ResearchBench evaluates the resulting papers and reviews through masked-node prediction.
- **ArtifactLinker** [5] ranks model–dataset links with GNNs before coding agents verify them. Its ArtifactBench supplies 14K artifacts and 51K relations for evaluation, with 96% audited accuracy.
- **IdeaForecastBench** [8] uses retrieved future papers to evaluate forecasts of new research ideas. Two judges apply this protocol to 624 rolling episodes across 52 research topics.

Multi-Agent & Agentic RL

- **CTM-AI** [9] coordinates specialist processors through competition and adaptive communication. This system achieves state-of-the-art results on MUStARD and UR-FUNNY, and 10+ point gains on StableToolBench and WebArena-Lite.
- **Multi-Agent Evolve** [10] co-evolves proposer, solver, and judge roles through RL self-play. Its Qwen2.5-3B-Instruct model gains 4.54% on average across math, reasoning, and knowledge benchmarks.
- **CoPE** [11] augments GRPO with multi-step action forecasting from successful rollouts. This yields up to 2.0× GRPO's sample efficiency on ScienceWorld without inference overhead.
- **Auto-Dreamer** [13] trains an offline memory consolidator with GRPO. Its agent outperforms the strongest baseline on ScienceWorld by 7 points while using 12× less active memory.

Selected Publications (* equal contribution)

[1] **H Yu**, Y Zhao, G Lin, J You. Building Social World Models with Large Language Models. *ICML 2026.*

[2] **H Yu***, Z Qi*, Y Zhao*, K Nottingham, K Xuan, B Majumder, H Zhu, P Liang, J You. SOTOPIA-RL: Reward Design for Social Intelligence. *Under review.*

[3] R Wang*, **H Yu***, W Zhang*, Z Qi*, M Sap, G Neubig, Y Bisk, H Zhu. SOTOPIA-π: Interactive Learning of Socially Intelligent Language Agents. *ACL 2024.*

[4] Y Zhao, B Liu, **H Yu**, Z Qi, S Wang, C Li, Y Liu, J You. Learning to Simulate Individuals from Macro Social Signals. *Preprint 2026.*

[5] **H Yu**, J You, P Clark, B Majumder, K Richardson. ArtifactLinker: Linking Scientific Artifacts for Automatic State-of-the-Art Discovery. *COLM 2026.*

[6] **H Yu**, K Xuan, F Li, K Zhu, Z Lei, J Zhang, Z Qi, K Richardson, J You. TinyScientist: An Interactive, Extensible, and Controllable Framework for Building Research Agents. *EMNLP 2025 Demo.*

[7] **H Yu***, Z Hong*, Z Cheng*, K Zhu*, K Xuan, J Yao, T Feng, J You. ResearchTown: Simulator of Human Research Community. *ICML 2025.*

[8] F Li, Z Tang, **H Yu**, Y Zhao, J You. Can Large Language Models Forecast What Researchers Study Next? *EMNLP 2026.*

[9] **H Yu***, Y Zhao*, L Blum, M Blum, P Liang. CTM-AI: A Blueprint for General AI Inspired by a Model of Consciousness. *NeurIPS 2026.*

[10] Y Chen, Y Wang, S Zhu, W Xu, **H Yu**, T Feng, M Zhang, M Patwary, J You. Multi-Agent Evolve: LLM Self-Improve through Co-evolution. *EMNLP 2026.*

[11] **H Yu***, Y Chen*, Y Zhao, X Ye, Y Liu, Z Hua, Y Xie, S Yang, J You. Co-Evolving Acting and Forecasting for Language Agents. *Under review.*

[12] **H Yu**, Z Xu, T Liu, A Ramakrishna, C Zhang, A Peshin, F Qi, G Lin, S Yang, K Rangadurai, J You, L Simon. Rethinking Agentic Continual Learning as Bayesian Discovery. *Under review.*

[13] C Ye, Y Liu, Y Wang, **H Yu**, Y Zhao, G Liu, J McAuley, J You. Auto-Dreamer: Learning Offline Memory Consolidation for Language Agents. *NeurIPS 2026.*